

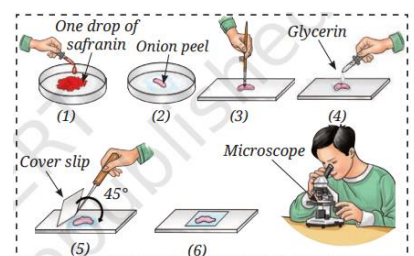
Chapter – 2: The Invisible Living World: Beyond Our Naked Eye

- Human eye – see obj. – above certain size
- Long time – tiny things – unknown – BUT – some time ago – people discover – curved piece – glass – small things – look bigger
- Piece of glass – shaped – lentil seed – thick – middle – thin – edge – called **lens**
- Over time – lenses – improved – become powerful
- New tools – magnifying glass to microscopes – help humans – see – eyes cannot
- Invention – microscope – open – fascinating hidden world – tiny living creatures
- **Explore** – some life forms – this chapter
- Already learnt – amazing variety – living beings
- Look around – beautiful plants, animals – all shapes, sizes, colours
- Some beings – very tiny – others – very big – differ – structure – BUT also – other features
- All beings – plants, animals – called **organisms**
- Think – how small – eyes can see
- Some people – reading glasses – how does it help? – OR – use magnifying glass – **observe** something?
- Activity –
 - Take – round-bottom flask – glass – fill water
 - Close mouth – flask – cork
 - Place flask – open book – look – letters
- Notice – interesting – letter – appear larger – flask – water – act – magnifying glass
- Use – real magnifying glass – obs. – small org. – ants – details – body – more clear
- Long time – people – curious – explore – tiny org. – BUT – cannot see – naked eyes
- **Interesting fact** –
 - 1665 – scientist – Robert Hooke – book – *Micrographia*
 - Hooke – careful observer, skilled artist
 - This book – detailed drawings – tiny things – never seen before – saw using tool – microscope
 - Microscope – things look – 200-300x bigger
 - 1 day – looking – thin slice – cork – made of – small, empty spaces
 - Compartments – remind him – honeycomb – drew – saw – called small space – cell
 - 1st time – cell word – used – science world – describe – basic unit – life
 - Same time – 1660s – Antonie van Leeuwenhoek – Dutch scientist – better lenses – allow him – more useful microscope
 - 1st person – see, describe – tiny beings – bacteria, blood cells – known – Father of Microbiology

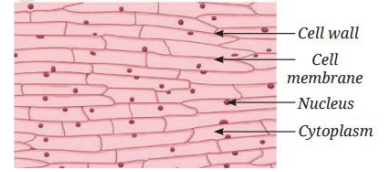


What is a Cell?

- All living beings – made of **cells** – wonder – looks – closer look – cell – microscope
- Activity –
 - Take – onion bulb – kitchen, garden – wash thoroughly – water
 - Cut onion – vertically – pieces
 - Take – 1 pc. – onion – pull – thin, transparent layer – inner surface – forceps – this layer – onion peel



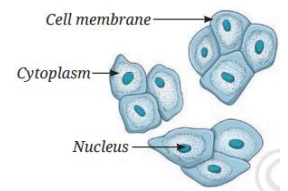
- Place peel – petri dish – few drops – safranin – 30 sec. – pink colour – cells – help see – clearly
- Thin brush – transfer peel – another dish – contain water – rinse peel – remove extra stain
- Place – stained peel – glass slide – thin brush – ensure – no break, fold
- Put – drop – glycerin – peel – prevent drying – improve clarity – better visuals
- Place – coverslip – peel – use needle – no air bubbles
- Blotting paper – wipe off – extra glycerin – edges – coverslip
- Obs. – slide – microscope – **compare** – given fig.



- Obs. – near rectangle structure – microscope
- These structures – cells – onion peel – close arrangement – no space – in b/w
- Try – obs. – peels – leaves – diff. plants – find out – all plants – cells

- Activity –

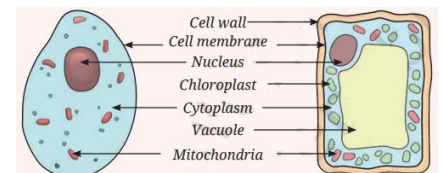
- Rinse mouth – clean water
- Use – blunt end – clean toothpick – scrape – inside cheek
- Place – scraped material – water drop – clean glass slide – spread even
- Add drop – methylene blue – slide – stain – improve visibility – increase contrast
- 1 min. later – drop glycerin – material – slide – prevent drying
- Place – clean coverslip – material – remove – excess glycerin – blotting paper
- Obs. slide – microscope – draw – vision



- Obs. – polygon-shape structure – cheek cells – inner lining mouth
- Similarities, differences – obs. – b/w cells – onion peel, human cheek
- Obs. – cells – 3 main parts
- Outer layer – **cell membrane** – round structure – middle – nucleus – covered – thin membrane
- Space b/w – cell membrane, **nucleus** – **cytoplasm**
- Some cells – onion peel – extra outer layer – **cell wall**
- Cell membrane – enclose – cytoplasm, nucleus – separate cells – porous – control – entry, exit – necessary, waste materials
- Cytoplasm – other components – compounds – carbs, proteins, fats, mineral salts
- Most life process – take place – cytoplasm
- Nucleus – regulate act. – within cell – also growth
- Cell wall – plant cell – rigidity, strength – plants – reason – all cells – compact arrangements – look firm

- **A step further**

- Cells – all plants – plant – rod-shape structures – **plastids**
- Some plastids – **chloroplasts** – chlorophyll – make green – help – photosynthesis
- Non-green parts – help – storage – substances
- Plant cells – large, empty space – **vacuole** – help store – imp. substance – get rid – waste – maintain shape – provide strength, support – plant
- Animal cells – vacuoles – gen. absent – if present – very small
- Small vacuoles – certain substances – dissolved – water
- Cell – bag of liquid – BUT also – complex structure – made – diff. parts – each – own special func. – cell – in turn – org. – work

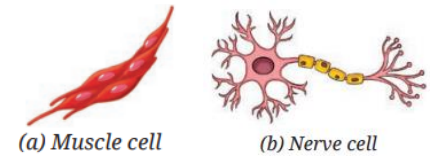


- Understand – basic structure – cell
- Also understand – plant, animal cells – differ – shape, structure

- Diff. animal cells – vary – shape, structure?

Variation in shape and structure of cells

- **Muscle, nerve** cells – human – given fig. – similarities, differences?
- Muscle cells – spindle shape – nerve cell – long, branched
- Some cells – round shape – others – long, thin
- No. of cells – vary – diff. org. – shape, structure – rel. to func.?
- Unique shape, size, structure – cells – help – specific func.
- Obs. – above act. – inner cheek cells – thin, flat – protective lining – inner surface – cheek
- Nerve cells – **neurons** – carry msg. – our body – elongated shape, branched structure – help reach – diff. parts – pass msg. quickly
- Plant cells – also vary – rectangular, elongated, oval, tube-like – some – long tubes – carry water
- Grade 7 – digestive sys. – diff. parts – diff. types – cells
- Grp. – muscle cells – food pipe – contract, relax – wave-manner – push food – stomach
- This movement – possible – muscle cells – thin, flexible, spindle-shape
- Stomach – diff. cells – diff. func. – muscle cells – stomach walls – churn food
- Other cells – inner lining – stomach – digestive juices, acid – help break – food
- All cells – work together – digestion possible



What Are the Levels of Organisation in the Body of a Living Organism?

- Body – living org. – complex way
- Cell – basic unit – life – like brick – basic unit – wall
- Grp. – similar cells – forms **tissue** – diff. tissues – organise – form **organ** – several organs – form **organ system** – major func. – all organ sys. – together – complete **organism** – plant, animals
- Levels – organisation – help understand – simple building blocks – form – complex living beings
- Life – complex living org. – begin – single cell – egg
- Egg – any org. – ability – divide repeatedly – form – complete living being
- Such beings – **multicellular organisms** – animals, humans, plants
- **Interesting fact** –
 - Yolk (yellow part – egg) – ostrich egg – largest known single cell – living world – 130-170 mm – diameter
 - Egg – extra non-cellular material – shell – protection – white liquid – nourishment – development

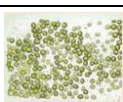
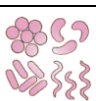
What are Microorganisms?

- Some living org. – just 1 or very few cells – so small – cannot see – naked eye – **microorganisms**
- Some – bacteria, *amoeba* – just 1 cell – unicellular – others – fungi, algae – many cells – multicellular
- Found – all around us – water, soil, air, inside us
- Their cells – look how? – are they similar – plant, animals?
- Obs. cells – microorganisms – use microscope – magnifies size – make them visible
- Scientists – low-cost, foldable paper microscope – not same level details – BUT – microscopic world – accessible
- Activity –
 - Take – container – collect pond, stagnant water – help – teacher, elder

- Use dropper – place drop – pond, stagnant water – microscope slide – put coverslip – obs.
- Obs. – tiny org. – pond, stagnant water
- Activity –
 - Take – beaker – collect – moist soil – nearby field, garden – do not touch soil – bare hands
 - Pour water – beaker – stir – glass rod – liquid – look dirty – fine particles – soil – soil suspension – keep aside – some time – mixture settle
 - Use dropper – take drop – water – top layer – place drop – microscope slide
 - Cover gently – coverslip – obs.
- Obs. – small org. – similar – above act.
- Indication – **soil suspension** – variety – tiny creatures – cannot see – unaided eye
- Tiny creatures – cannot see – unaided eye – **microbes**
- Activity –
 - Grp. – students – Grade 8 – above act. – also collected – info. – library, internet
 - Record data – diff. tables – identified microorganisms – protozoa, algae, fungi, bacteria
- Org. – pond water –

S. No.	Microorganisms	Recorded data
1.	<i>Amoeba</i> (Protozoa) 	Single cell, moving, irregular shape
2.	<i>Paramecium</i> (Protozoa) 	Single cell, moving – specialized structure
3.	Algae 	Singel cell, looks green – green pigment, movement – specialized structures

- Org. – soil suspension –

S. No.	Microorganisms	Remarks
1.	Bread Mould (Fungi) 	Branched filament – no chlorophyll – sac-like structure
2.	Mould (Fungi) 	Branched filament – no chlorophyll – brush-like structure
3.	Algae 	Spherical – chlorophyll present
4.	Bacteria 	Spherical, comma, spiral, rod – shapes – 1 long hair-like structure – many small hair-like structure

- Also obs. – these microorganisms – OR – anything diff.? – record it
- Microorganisms – everywhere – only visible – microscope – device – magnify – 100-400x
- **Interesting fact** –
 - **Viruses** – microscopic, acellular – multiply – enter living cells
 - Infect – plants, animals, bacterial cells – cause disease

How Are We Connected to Microbes?

- Microorganisms – other places?
- Ever seen – orange, tomato, other foods – rot – left outside – some time?

- Notice – powder, cotton-like growth – reason – infected – microbes
- Where does microbes come from?
- Reason – microbes – found everywhere – water, soil, air, food items
- Use microscope – explore surface – leaves, stems, roots, other parts
- Like plants, animals – microbes – great diversity
- Some found – extreme climate conditions – hot water springs – snow cold zones – moderate temp.
- Already know – some org. – inside us – even guts
- Grade 7 – *curiosity* – chapter – life processes in animals – learnt – intestine – many bacteria – help in digestion
- Like plants, animals – microbes – vary – shape, size, structure – spherical, rod-shape, irregular

Key players in cleaning the environment

- Activity –
 - Take – empty container – fill halfway – garden soil
 - Add – fruit, vegetable peels – container – put layer – soil – leave aside
 - 2-3 weeks later – obs. changes – taken place
 - Obs. – any diff. – contents – container?
- You find – peels – fruits, vegetables – turned – dark-colour material – **manure**
- Manure – rich – nutrients – help increase – fertility of soil
- Above act. – soil – various microbes – some – fungi, bacteria – act – plant waste – break down – simple, nutrient-rich manure
- Obs. – gardeners – school, fields – collect dry leaves, plant waste – pits – make natural manure
- Look around – decaying plants, fallen leaves – container, garden – disappear – sometime later
- Microbes – break down – complex substance – fallen leaves – simpler substance – rich – nutrients – process – **decomposition**
- Nutrients – go back – soil – help plants – better growth
- Microbes – also decompose – dead animals
- Microbes – recycle waste – return imp. nutrients – nature
- Manure formation – optimal temp. – appropriate moisture levels
- By now – understand – bacteria, fungi – microbes – imp. role – our lives – also decompose – animal wastes
- Above act. – infer – microbes – help – plant growth – BUT also – clean env. – break down waste
- **A step further**
 - Microbes – src. – **biogas**
 - Many microbes – bacteria, fungi – live – oxygen-free env. – some – ability – decompose – plant, animal waste
 - During process – release mixture – gases – biogas – composition – CO₂, CH₄
 - This gas – fuel src. – cooking, heating, elec., vehicles

Microorganisms and food

- Activity –
 - Take 2 bowls – A, B
 - Each – 200 g flour (*atta, maida*) – add pinch – sugar
 - Bowl A – small amt. – yeast powder – mix well
 - Bowl B – no yeast – compare results
 - Knead flour – 2 bowls – warm water – soft dough
 - Cover dough – damp cloth – keep it – warm place

- Obs. – both bowls – 4-5 hrs. later
- Sometime later – notice – dough – bowl A – risen, fluffy – diff. smell than – bowl B
- Role of yeast? – why add – sugar, warm water?
- Yeast – microbe grp. – **fungi** – grows well – warm conditions
- Grade 7 – chapter – life processes in animals – recall – yeast – respire – break food – release energy – growth, life processes
- This process – CO₂ released – forms bubbles – dough – fluffy, soft
- Yeast – also produce – small amt. – alcohol – diff. smell
- Special prop. – yeast – make breads, cakes, more
- Addition to yeast – bacteria – *lactobacillus* – help – fermentation – batter – make idli, dosa – dough – make *bhatura*
- Activity –
 - Take – 2 small glass bowls – A, B
 - Pour – bowl A – lukewarm milk – bowl B – cold milk
 - Add – small spoon – curd – each bowl – mix well – spoon
 - Cover both bowls – bowl A – warm place – bowl B – cool place – few hrs., overnight
 - Obs. changes – glass bowls – note predictions, obs.
- Obs. – bowl A – milk – turned – curd – few hrs. later – little sour
- Bowl B – milk – not curdled – BUT – little sour
- Curd – several bacteria – 1 – *lactobacillus* – feeds on sugar (lactose) – multiply – ferment milk – form curd
- Unlike yeast – bacteria – no alcohol – BUT – lactic acid – makes curd sour
- These bacteria – grow well – warm conditions – reason – bowl A – curd present – bowl B – no curd
- Categorize microbes – diff. types – protozoa, fungi, bacteria, algae, etc.
- Some bacteria – *Rhizobium* – form swollen regions – nodules – live there
- Roots – certain legumes – beans, peas, lentils – nodules – *rhizobium*
- This bacteria – trap N₂ – air – make useful for plants
- Help plants – grow better – no chem. fertilisers – reason – farmers grow legumes – rotation – other crops – natural increase – N₂ – soil healthy – next crop

Amazing microalgae: tiny helpers in water

- Microalgae – microscopic plant-like org. – live – water, soil, air, trees
- Make food – use sunlight – this process – also release O₂ – produce – 50% + O₂ supply
- Rich – nutrients – serve – food src. – aquatic animals
- Some – *spirulina*, *chlorella*, diatoms – used – health supplements, medicines
- Also help – clean water – used – biofuel
- BUT – pollution, climate change, habitat destruction – threaten – microalgal diversity, abundance
- Imp. – conserve – tiny org. – protect env. – maintain O₂ balance – earth
- **Interesting fact** –
 - *Spirulina* – microalga – superfood – health benefits
 - Also – good src. – vitamin B₁₂ – essential – our body
 - Rich – protein – 60% + - body weight – only – small amt. – fat, sugar
 - Farming – *spirulina* – feasible livelihood opp. – grow – easy steps –
 - Set – clear glass tank – bright place – no direct sunlight
 - Cover tank – shade net – keep tank – moderate temp. conditions
 - Fill tank – pond water
 - Add – live *spirulina* – collect – pond



- Stir – growing microbe – 2x / week
- 3-6 weeks later – *spirulina* – harvested – filter it – fine cloth
- Conservation – microalgae – good practice – ensure – food security – livelihood development

Why is Cell Considered to Be a Basic Unit of Life?

- Body – all living org. – tiny building blocks – cells
- Single cell – various components – various func.
- Bodies – all plants, animals – many cells – **multicellular** org.
- These org. – cells – specialized individual func. – BUT also – cooperation – better chance – survival
- Some microbes – bacteria, protozoa – just 1 cell – **unicellular** org.
- All func. – necessary – survival – single cell
- Other microbes – algae, fungi – 1 or more cells – ex. – yeast – single cell – BUT – mould – multi cells
- Like animal, plant cells – microbe cells – also – cell membrane
- Fungi cells – additional cell wall – BUT – no chloroplast – no photosynthesis
- Bacteria – no well-defined nucleus, nuclear membrane – instead – nucleoid – distinguish them – yeast, protozoa, algae, fungi, plants, animals
- Only looked – basic cell structures – other components – study – higher classes
- Obs. – subcellular components – microscopes – high magnification
- Electron microscope – magnify 10,00,000x – see – more structures present
- By now – understand – all living beings – incl. microbes – 1 or more cells
- Cells – differ – size, shape, structure – plant, animal cells – some diff. – understand diff. – help learn – org. func. differently
- This chapter – learnt – beneficial microbes – BUT - some microbes – harmful – plants, animals – incl. humans – learn – next chapter

